

My project is an interactive computational art piece developed using p5.js that explores how code constrains player behavior within a game. It draws inspiration from the anime <Cells at Work> as well as the texts *By Any Means Necessary* and *Travesty Generator*. I was particularly struck by the section in the *Travesty Generator* describing "I," "can't," "breathe," and "I can't breathe." I wondered if I could express that feeling in a different form. That is how I came to create this project, using code to place constraints on the player's actions. Players take on the role of a red blood cell inside the body of the individual involved in the "I Can't Breathe" incident, tasked with transporting oxygen. Although the body is destined to die from oxygen deprivation, the red blood cells within continue to strive to deliver oxygen to every part of the system. In the game, the player navigates through blood vessels as a diligent red blood cell, collecting and transporting oxygen bubbles. However, this is not a traditional game about collecting points for a high score. Through coded constraints, the game ensures that every effort the player makes is doomed to fail, illustrating how computational rule systems can embody the ruthless and irreversible nature of systemic oppression.

Visually, the project centers on a beating dark-red heart and an active ECG trace, accompanied by the muffled sound of a heartbeat that foreshadows a tragic conclusion. The underlying code is designed to evoke the sense of failure and despair inherent in the fact that the game cannot be won. First, I implemented specific timing mechanics: during the initial 30 seconds of gameplay, oxygen is continuously generated, creating an illusion of normal operation and steady rhythm. At the 30-second mark, oxygen generation ceases signaling that the body's host can no longer breathe. While the player can continue to play, the outcome is already sealed.

Second, I incorporated real-time sound effects to reflect the body's health status. During the two-minute countdown, the sound of a rapid ECG monitor plays during the final 26 seconds, heightening tension for the player. This signals the body's gradual decline toward death following prolonged oxygen deprivation and the exhaustion of internal oxygen reserves. This mirrors the infinite loop found in *Travesty Generator* ; no matter how hard the player tries, they cannot save the host—much like the system crash caused by an infinite loop in *Travesty Generator* . Furthermore, while players appear to possess full

freedom of action—such as WASD movement, free lane-switching, and interactive pickups—the underlying code creates the illusion that they might still save the body's original owner. In reality, the game's core variables are hard-coded and locked; there are no branching paths that allow for victory or a reversal of fate. This design forces players to viscerally experience a sense of powerlessness against the cold, programmed constraints of algorithms and code—mirroring the helplessness of the real-life figure at the heart of this event, who was unable to breathe while under police pressure.

In essence, this project employs code logic as an artistic constraint to represent oppression: players may attempt a rescue, but their efforts are destined to be in vain. This effectively illustrates the concept of "Code as Constraint."

Work Cited

Audio download websites: https://sounddino.com/zh/effects/heart-and-cardiogram/#group_1654
Source of the red blood cell image: <https://yapability.itch.io/blood-cell-inspired-organism-animation>
AI conversation link: <https://share.gemini.google/sTKzi4Mh3OBC>